

ANANTHU N

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Agentic AI, RAG systems, and Federated Learning. Experienced in LLM orchestration (LangChain, LangGraph), backend API development (FastAPI), and production ML pipelines using Python and PyTorch. Published a Springer conference paper on federated healthcare AI.

TECHNICAL SKILLS

Programming & Databases: Python, SQL, MySQL

Machine Learning & Deep Learning: Scikit-learn, PyTorch, PEFT, LangChain, Transformers

NLP & LLMs: Hugging Face Transformers, Sentence-BERT, Embeddings, Text Classification, Summarization, RAG

Knowledge Graphs: RDF, SPARQL, Blazegraph, KG-RAG, Entity Linking

Deployment & MLOps: Docker, FastAPI, Git/GitHub, MLflow, DVC

EDUCATION

MSc in Data Science — University of Kerala

2023 – 2025 — CGPA: 8.3/10

BSc in Statistics — University of Kerala

2020 – 2023 — CGPA: 7.9/10

EXPERIENCE

Project Assistant — University of Kerala

Sep 2025 – Present

Project: "Enhancing Network Analysis with Machine Learning: Application in Node Classification, Link Prediction, and Community Detection"

- Developed a privacy-preserving Federated Graph Learning framework (FedGraph, PyTorch Geometric) for link prediction over distributed patient-disease knowledge graphs across simulated hospital clients, using synthetic EHR data (~185K patient-disease pairs via Synthea); achieved 0.815 AUC and 99.45% hit rate@2 after 4 federated training rounds — published as a Springer conference paper (AI4S 2025).
- Built a federated query generation framework over the same distributed healthcare data sources, fine-tuning defog/llama-3-sqlcoder-8b with LoRA and implementing a SentenceTransformers + FAISS RAG pipeline for schema-aware NL-to-SQL generation across heterogeneous database schemas.

AI Intern — ZenturioTech, Technopark Trivandrum

Apr 2025 – Jun 2025

- Developed a transformer-based expense classification pipeline and generated synthetic expense data using large language models to address data sparsity and class imbalance, improving model generalization across low-frequency categories.
- Implemented an end-to-end inference workflow and deployed the expense classification model via a Streamlit application, enabling real-time predictions and structured evaluation.

ACHIEVEMENTS

- Publication:** Published Springer conference paper *Federated Learning over Cross-Hospital Knowledge Graphs for Privacy Preserving Disease Prediction* in **AI4S 2025**.

PROJECTS

Autonomous Market Intelligence Agent (Agentic AI System)

- Engineered and orchestrated a multi-agent AI system using Python, LangGraph, and LLaMA-3 (Groq) to autonomously perform competitive market analysis, including research planning, web data acquisition, structured extraction, feature normalization, comparison, scoring, and report synthesis.
- Converted raw web information into clear competitive insights, including feature differences, pricing patterns, market gaps, risks, and recommendations, with validation logic to ensure accurate and reliable analysis.

RefineCap: Parameter-Efficient BLIP Enhancement for Image Captioning

- Engineered RefineCap, a 5-stage visual refinement module — Residual MLP, Q-Former-style learnable query cross-attention, importance-weighted semantic attention, a learnable fusion gate, and an auxiliary caption-length head — inserted between BLIP's vision encoder and text decoder via non-invasive forward-pass patching, and fine-tuned it by selectively unfreezing only the final decoder layers and language modeling head while keeping the base model frozen.
- Cut hallucinated captions via CLIP-reranked beam-search candidates, lifting CIDEr +59.67%, BLEU-4 +47.88%, ROUGE-L +22.62%, and METEOR +17.60% over base BLIP on Flickr30k.

Self-Healing RAG — Autonomous Research Assistant Platform

- Architected a stateful, self-healing RAG pipeline on LangGraph (FastAPI, Python), fusing Qdrant dense retrieval (BAAI/bge-large-en-v1.5) with BM25 via Reciprocal Rank Fusion and cross-encoder reranking (ms-marco-MiniLM-L6-v2), then layered Corrective RAG (CRAG) on top with live arXiv paper acquisition, sentence-level grounding evaluation, and a 7-type failure taxonomy driving automated query-rewrite retries via LLaMA-3.3-70B (Groq).
- Reached 78% precision@10 (+16% over baseline), cut hallucinated claims by ~30%, resolved >90% of failing queries within two retries, and containerized the pipeline with Docker at >90% test coverage for production deployment.

Deterministic Knowledge-Grounded RAG Engine for Clinical Safety

- Architected a deterministic Knowledge Graph-grounded RAG QA pipeline (Python, PyTorch, Hugging Face Transformers, FAISS, Snowflake Arctic embeddings) enforcing SPARQL-verified drug-interaction facts against Blazegraph RDF triplets before LLM rendering, eliminating hallucinated outputs via a grounding-guard and deterministic fallback policy layer.
- Deployed a production FastAPI inference service, Dockerized with hardware-aware dual-model loading (Phi-3-mini / TinyLlama-1.1B, CPU/CUDA fallback via psutil and PyTorch device detection), SentenceTransformer entity linking (MiniLM-L6-v2 + FAISS index), and BM25-hybrid reranked retrieval over a Wikipedia drug corpus.